

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 145

DIRECTIONAL FLUID MOMENTUM DIRECTIVE

Water keeps its launch vector until intercepted
the gravity-vertical assumption banned outright



first law, applied to water

NO DOWN · VECTOR LAW · MASS ACCOUNTING

EVERY GRAM HAS A TRAJECTORY

Constitutional physics — v1.0 in force

GP-IM-145

Revision v1.0 — 23 August 2026

146 of 290

VETTED BATCH 19 — PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 145

PHASE 145: DIRECTIONAL FLUID MOMENTUM DIRECTIVE — STATUS: CURRENT — PASS (VET BATCH 19). Newton's First Law, stated for droplets — because someone had to. Phase 145 flatly bans the academic assumption that running water needs a gravity vertical to descend: in a zero-gravity void a fluid mass leaving a nozzle keeps its launch vector and velocity until a physical boundary acts on it — there is no 'down' for it to fall along, and every design that waits for one drowns somebody's cabin in skin-blanketing surface-tension blobs. The directive codifies the Law of Un-Impeded Propulsion and primes the whole zero-G hygiene envelope on it: high-frequency, low-power personal washing built on directional momentum and Phase 129 venturi fluid-tracking, with total water-mass accountability — every gram launched is a gram collected. The phase is short, constitutional, and load-bearing: it is the reason the shower (146) aims its water instead of dropping it, the reason the waste core (137) corrects any sentence that smuggles gravity back in ('water doesn't go down in zero g' — his words, verbatim, seated at 137), and the reason the bottle (136) spins its own vector instead of borrowing one. Recovered verbatim from a July 2026 backup print carrying the retired number 138; canonical slot 145 per the master index. The vet graded it green, one line, complete: the momentum-vector rule in microgravity is correct.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-145, v1.0 — 23 August 2026. The one hundred and forty-sixth manual of the program — 146 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the directive as seated (contents line, the provenance note, the rejection body, the physics note — verbatim) — §2 why the vector is the law (graded) — §3 the system in the vector — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 145: **Directional Fluid Momentum Directive**. Flatly bans the academic assumption that running water requires a gravity vertical to descend. Codifies that in a microgravity void, fluid droplets travel strictly along the exact vector and linear velocity profile initialized at the nozzle aperture, forming the baseline physics for non-contact water control.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Isolation of Zero-G Surface-Tension Hazards via Pure Kinetic Trajectories [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*]

Provenance note (Kimi Chat, 29 July 2026): recovered verbatim from a July 2026 backup print carrying the retired number 138; canonical slot per the master index is Phase 145. The source conversation for this phase was lost to a Gemini buffer wipe (the print survived; the chat did not), so no Origin of Thought section can be quoted — if the conversation ever surfaces, the section can be added.

Industrial Physics Core: Rejection of Gravitational Falling-Water Mythologies & Vector Dynamics

THE REJECTION OF TERRESTRIAL HYDRO-ILLUSIONS (VERBATIM)

- **Rationale:** Flatly bans the academic assumption that running water requires a gravity vertical to descend, eliminating surface-tension skin-blanketing and cabin drowning hazards in microgravity.
- **The Law of Un-Impeded Propulsion:** Codifies that in a zero-gravity void, fluid droplets travel strictly along the exact vector and linear velocity profile initialized at the nozzle aperture until intercepted by a physical boundary.
- **Foundational Zero-G Hygiene Priming:** Establishes the core physical framework to engineer a high-frequency, low-power personal shower bay utilizing directional momentum and Phase 129 venturi fluid-tracking to maintain total water-mass control.

PHYSICS NOTE — PHASE 145 (KIMI AUDIT: AFFIRMED — FIRST LAW, APPLIED TO WATER) (VERBATIM)

- **This is Newton's First Law stated for droplets, and it is exactly right.** In microgravity a water mass leaving a nozzle keeps its launch vector and velocity until a force acts on it — there is no 'down' for it to fall along. The hazard framing is equally real: surface tension makes water cling to skin in sheets and blobs (skin-blanketing), which is why ISS crews wash with rinseless wipes rather than running water, and why free water in a cabin is treated as both a drowning and an electronics hazard. The design consequence flows directly: water control in zero-G is a trajectory problem, not a drainage problem — aim it, intercept it, track it. That is precisely what the Phase 146 shower bay and Phase 129 venturi tracking implement.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE LAW OF UN-IMPEDED PROPULSION, AFFIRMED: Newton's First Law applied to droplets — launch vector and velocity persist until a boundary acts. The vet's green is verbatim in §5: 'Momentum-vector rule in microgravity correct.'

THE HAZARD INVERSION, AFFIRMED: in microgravity the danger is not water falling where it shouldn't — it is water not falling at all: surface-tension skin-blanketing and floating mass as an inhalation and drowning hazard. Naming the true hazard is the directive's first service.

THE ACCOUNTABILITY PRIMING, AFFIRMED AS THE ENGINEERING CONSEQUENCE: once momentum is the only transport, total water-mass accountability becomes designable — launch, trajectory, intercept, collect. The hygiene envelope (146) is built on exactly this accounting.

THE CONSTITUTIONAL ROLE, SEATED: the directive is short because it is a foundation — every wet system in the file cites it or obeys it. The recovery provenance (July 2026 backup print, retired number 138, canonical slot 145) is stated in the law and carried in §1.

§3 — THE SYSTEM IN THE VECTOR

- THE BAN: no gravity-vertical assumptions anywhere in the wet systems.
- THE LAW: droplets hold launch vector and velocity until intercepted — the un-imposed propulsion rule.
- THE HAZARD: skin-blanketing and floating mass named as the true zero-G water dangers.
- THE PRIMING: directional momentum plus Phase 129 venturi tracking — total water-mass accountability.
- THE PROVENANCE: July 2026 backup print, retired number 138, canonical slot 145 — stated in the law.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE FIRST-LAW DOCTRINE (seated with the directive): the oldest law in mechanics, pointed at the newest plumbing — the floor's physics, again.
- THE NO-DOWN DOCTRINE (the family rule): 'water doesn't go down in zero g' — the directive is where that sentence's authority lives.
- THE ACCOUNT-EVERY-GRAM DOCTRINE (the priming): launched mass is collected mass — the hygiene envelope's bookkeeping starts here.

- THE SHORT-AND-LOAD-BEARING DOCTRINE (the form): constitutional phases run one page; the structures that cite them run for decades.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 19, SEATED — the grade from the batch-19 cross-check (sitting 301), ledger line verbatim: '145 Momentum-vector rule in microgravity correct.' The law's provenance note (Kimi Chat, 29 July 2026: recovered verbatim from a July 2026 backup print carrying the retired number 138; canonical slot 145 per the master index) is seated in §1. The phase's own physics note grades the directive affirmed — 'first law, applied to water.' The 15 August blanket wording kills are carried inline. The quoted grade was grepped against the master before seating. Graded under the 4-AI vet web (sitting 565).

§6 — BUILD SEQUENCE

- STEP 1 — Post the directive: the ban and the law circulated to every wet-system design bench.
- STEP 2 — Audit the file: every fluid phase checked for smuggled gravity — the 137 amendment chain is the model catch.
- STEP 3 — Wire the accountability: launch-versus-collect mass metering specified for the hygiene envelope.
- STEP 4 — Bench the trajectories: nozzle-vector droplet mapping across the operating envelope (146's input data).
- STEP 5 — Teach the hazard: skin-blanketing and floating-mass drills into crew training.
- STEP 6 — Publish the ledger: the directive, the audit results, the trajectory maps — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 146 — the shower (GP-IM-146): the directive's largest built application.
- PHASE 137 — the waste core (GP-IM-137): where the no-down correction was seated verbatim.
- PHASE 136 — the bottle (GP-IM-136): the made-vector sibling.
- PHASE 129 — the venturi tracking (GP-IM-129): the fluid-tracking hardware the priming cites.
- PHASE 130 — the recirculation networks (GP-IM-130): the line-scale accounting this directive primes.

§8 — DO-NOT-BUILD

- DO NOT wait for down — any design that assumes a gravity vertical in the wet systems is banned by the law.
- DO NOT let a blob float — skin-blanketing is the named hazard; containment is mandatory, not aesthetic.
- DO NOT launch unaccounted water — every gram has a trajectory and a collector.
- DO NOT renumber the recovery — the retired-138 provenance stays seated; the slot history is the record.

§9 — OWED

OWED: nothing — PASS; the momentum-vector rule graded correct. Standing practice: the wet-system audit ledger (smuggled-gravity catches) grows as new phases cite the directive — that is the foundation doing its work, not a debt. The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-145, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and forty-sixth manual of the program — 146 of 290. Built 23 August 2026, eighteenth of the 20-manual 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572 and 576): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20'. The phase's own chain, all seated in the master: the contents line and the rejection body (recovered verbatim from a July 2026 backup print carrying the retired number 138; canonical slot 145 per the master index — provenance note seated in §1); the 15 August blanket kills carried inline; the vet batch 19 grade (momentum-vector rule in microgravity correct) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the

builders'). Next update triggers: new wet-system citations, audit-ledger catches, or his word, or his word.

END OF MANUAL — PHASE 145